

EXPLANATION OF SQL PRODUCT PM

This SQL Product PM assignment focused on creating and analyzing a simple product analytics database using SQL. Two main tables were created: `users` and `conversions`. The `users` table stored product user information such as ID, name, and engagement score. The `conversions` table stored conversion data for users. Sample data was inserted into both tables to simulate a product management and analytics environment.

A LEFT JOIN was used to combine the `users` and `conversions` tables. This join was implemented through a SQL VIEW named `user_view`. The LEFT JOIN ensured that all users from the `users` table appeared in the output, even if some users did not have conversion data. Users without matching conversion values displayed NULL values in the conversion column. This type of join is useful in product analytics because companies often want to analyze all users, including inactive or non-converting users.

The query `SELECT * FROM user_view;` displayed the combined user and conversion data. This helped visualize how joins work across multiple tables. Another query used `GROUP BY` along with the aggregate function `COUNT()` to count how many users belonged to each conversion group. This demonstrated how SQL can summarize product metrics and categorize user behavior.

The assignment also used aggregate functions such as `MAX()` and `MIN()` on the conversion column. The MAX function identified the highest conversion value, while the MIN function identified the lowest conversion value. Aggregate functions are important in analytics because they help summarize large amounts of data quickly and support data-driven decisions.

A subquery was also practiced using an UPDATE statement. The query reduced engagement scores by 10 for users whose engagement value was 120. The subquery selected IDs of users meeting the condition, and the outer UPDATE query modified only those rows. This demonstrated how subqueries can be used with UPDATE statements to perform targeted data modifications.

During the assignment, debugging and syntax correction were also practiced. Issues such as incorrect parentheses, missing commas, case-sensitive text matching, and improper subquery conditions were identified and fixed. Trial-and-error learning improved understanding of SQL syntax and logical thinking.

This assignment aligns strongly with product analytics and investigative skills. SQL allows analysts and product managers to explore user engagement, identify high-performing users, and understand conversion behavior. Queries involving joins, aggregate functions, views, and subqueries are commonly used in real-world dashboards and reporting systems.

Overall, this project strengthened practical SQL skills including table creation, inserting records, joins, views, grouping, aggregation, subqueries, and updates. These skills are essential for data analyst and product management roles. Regular practice with SQL also supports portfolio development and preparation for future internships and high-paying technology careers.

DB FIDDLE LINK ☐

<https://www.db-fiddle.com/f/2fJFhncnHVfRqSkAXZe7gb/0>